AgriScore

Location- Natungram, Dist. Hooghly, Mogra, West Bengal, 712148 Mar 29, 2026

SAMPLE INFORMATION

Sample Depth: 0-20 cm (Topsoil)

Land Use: Agricultural (Crop Cultivation)

Previous Crop: Rice (Kharif Season)

FIELD CHARACTERISTICS

Parameter	**Value**
Total Land Area	4,086 square metres (3.05 Bigha)
Cultivable Area	3,700 square metres (2.76 Bigha)
Soil Type	Alluvial Loam
Soil Color	Dark Brown
Soil Texture	Sandy Loam
Drainage Status	Well Drained

Table 1: Field and physical characteristics of the sampled land

SOIL TEST RESULTS

Primary Chemical Parameters

Parameter	Test Result	Optimal Range	Rating
pH (1:2.5 soil-water ratio)	6.8	6.5 - 7.5	Optimal
Electrical Conductivity (EC)	0.42 dS/m	<1.0 dS/m	Normal
Organic Carbon (OC)	0.68%	0.5 - 0.75%	Medium

Table 2: Primary soil chemical parameters and their ratings

Macronutrients (NPK)

Nutrient	Available Content	Optimal Range	Rating
Nitrogen (N)	245 kg/ha	280 - 450 kg/ha	Low
Phosphorus (P)	28 kg/ha	22 - 56 kg/ha	Medium
Potassium (K)	185 kg/ha	140 - 280 kg/ha	Medium

Table 3: Macronutrient availability and classification

Secondary Macronutrients

Nutrient	Available Content	Optimal Range	Rating
Calcium (Ca)	1680 ppm	1400 - 2000 ppm	Adequate
Magnesium (Mg)	142 ppm	100 - 200 ppm	Adequate
Sulfur (S)	11.5 ppm	10 - 20 ppm	Medium

Table 4: Secondary macronutrient status

Micronutrients

Micronutrient	Available Content	Optimal Range	Rating
Iron (Fe)	14.2 ppm	10 - 20 ppm	Sufficient
Manganese (Mn)	7.8 ppm	8 - 11 ppm	Low
Zinc (Zn)	1.2 ppm	1 - 3 ppm	Sufficient
Copper (Cu)	0.85 ppm	0.8 - 1.0 ppm	Sufficient
Boron (B)	0.52 ppm	0.5 - 1.0 ppm	Sufficient

Table 5: Micronutrient status in soil sample

Soil Physical Parameters

Parameter	Test Result	Optimal Range	Rating
Soil Moisture Content	18.5%	15 - 25%	Adequate
Water Holding Capacity	42%	35 - 50%	Good

Table 6: Physical properties of the soil sample

Particle Size Distribution

Particle Type	Percentage (%)	Classification
Sand (2.0 - 0.05 mm)	58%	Sandy
Silt (0.05 - 0.002 mm)	28%	Loam
Clay (<0.002 mm)	14%	-
Textural Class	**Sandy Loam**	

Table 7: Soil particle size distribution and textural classification

INTERPRETATION AND RECOMMENDATIONS

Soil Health Assessment

The soil analysis indicates that the tested land has **moderately fertile soil** with the following characteristics:

- **pH Level:** The soil pH of 6.8 is within the optimal range for most agricultural crops, indicating neutral soil conditions that support good nutrient availability[1].
- **Salinity Status:** The EC value of 0.42 dS/m indicates normal salinity levels with no salt stress for crop growth[2].
- **Organic Matter:** Medium organic carbon content (0.68%) suggests moderate soil fertility. Increasing organic matter through farmyard manure or compost is recommended[3].
- **Nutrient Status:** Nitrogen levels are low, while phosphorus and potassium are in the medium range, requiring targeted fertilizer application.

Fertilizer Recommendations

Based on the soil test results, the following fertilizer doses are recommended for the upcoming **Rabi season wheat crop** (per acre):

Nutrient	Recommended Dose	Fertilizer Source	Application Method
Nitrogen (N)	60 kg/acre	Urea (130 kg/acre)	Split: 50% basal + 25% tillering + 25% grain filling
Phosphorus (PO)	25 kg/acre	DAP (54 kg/acre)	Full basal application
Potassium (KO)	20 kg/acre	MOP (33 kg/acre)	Full basal application
Zinc (Zn)	2 kg/acre	Zinc Sulfate (10 kg/acre)	Basal application

Table 8: Crop-specific fertilizer recommendations for wheat cultivation

Soil Management Practices

- **Organic Matter Enhancement:** Apply 5-6 tonnes of well-decomposed farmyard manure (FYM) or compost per acre before land preparation to improve organic carbon levels[4].
- **Micronutrient Management:** Foliar application of manganese sulfate (0.5% solution) at 30 and 45 days after sowing to address low manganese levels.
- **Moisture Management:** Maintain soil moisture at field capacity during critical growth stages. The current moisture level of 18.5% is adequate for sowing.
- **Soil Conservation:** Practice crop rotation with legumes (pulses) to enhance nitrogen fixation and improve soil structure.
- **pH Maintenance:** Monitor soil pH annually. If pH drops below 6.0, apply agricultural lime at 200-400 kg/acre.

Crop Suitability

Based on the soil characteristics, the following crops are **highly suitable** for this land:

- **Cereals:** Wheat, Rice, Maize
- **Pulses:** Chickpea, Lentil, Green Gram
- **Oilseeds:** Mustard, Sunflower, Groundnut
- **Vegetables:** Potato, Tomato, Cabbage, Cauliflower

REMARKS

The soil sample was representative of the field and collected following standard sampling procedures. Fertilizer recommendations are based on expected crop yield targets and should be adjusted based on actual crop performance and local agronomic practices. For detailed crop-specific guidance, consult with local agricultural extension officers.